CHAPTER 1

Complex numbers: the algebraic point of view

Computing in $\mathbb{C}$, conjugation, binomial formula

Objectives. Compute with the algebraic form; use the conjugate to obtain an inverse or a quotient; solve a linear equation, or a simple equation involving z and $\bar{z}$; expand a power with the binomial formula.

Prerequisites. Algebraic manipulation and standard identities; quadratics; binomial coefficients and Pascal's rule (specialty course); proof by induction.

Fluency

Fluency · No calculator, five minutes.

1. Expand $(2-3x)^2$.
2. Solve in $\mathbb{R}$ the equation $x^2-4x+13=0$.
3. Expand $(a+b)^3$.
4. Give the value of $\binom{5}{2}$.
5. Write $\frac{1}{\sqrt{2}-1}$ without a square root in the denominator.
6. Solve in $\mathbb{R}$ the equation $3x-5=2x+7$.